TT-Geometry of $\mathrm{Ver}_{2^3}^{+}$

Geometric and Homological Methods in Representation Theory
Kennesaw State University

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October 10, 2026

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Finite Tensor Categories

Definition (Finite Tensor Categories)

A k-linear abelian rigid monoidal category $\mathcal{C}$ is a **finite tensor category** if

1. $\mathcal{C} \simeq \Lambda - \text{mod}$ for a finite dim k-algebra Λ
2. A bilinear biexact tensor product $- \otimes - : \mathcal{C} \times \mathcal{C} \to \mathcal{C}$
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Theorem (Deligne)

Over a an algebraically closed field $\text{char}\, k = 0$, *any symmetric tensor category (of moderate growth) admits an exact, faithful, symmetric tensor functor to* sVec_k.

Higher Verlinde Categories

Definition

Let k be an algebraically closed field of characteristic p.

$$\mathrm{Ver}_{p^n} := \mathcal{A}_{n,p} - \mathrm{mod} \qquad \mathrm{Ver}^+_{p^n} := \mathcal{A}^+_{n,p} - \mathrm{mod}$$

$$\mathcal{A}_{n,p} = \prod_{i,j=p^{n-1}-1}^{p^n-2} \mathrm{Hom}_{\mathbf{SL}_2(k)}(T_i, T_j)^{\mathrm{op}} \quad \mathcal{A}^+_{n,p} = \prod_{\substack{i,j=p^{n-1}-1 \\ i,j \text{ even}}}^{p^n-2} \mathrm{Hom}_{\mathbf{SL}_2(k)}(T_i, T_j)^{\mathrm{op}}$$

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Example ($\mathrm{Ver}^+_{2^3}$)

$$\mathcal{A} := \mathcal{A}^+_{3,2} = \prod_{i,j\in\{4,6\}} \mathrm{Hom}_{\mathbf{SL}_2(k)}(T_i, T_j),$$

Higher Verlinde Categories

Definition

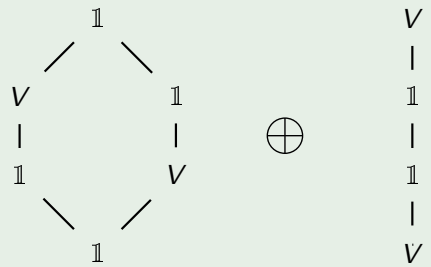
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Example ($\mathrm{Ver}^+_{2^3}$)

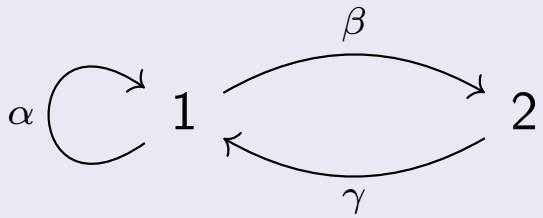
$$\mathcal{A} := \mathcal{A}^+_{3,2} = \prod_{i,j\in\{4,6\}} \mathrm{Hom}_{\mathbf{SL}_2(k)}(T_i, T_j), \qquad {}_{\mathcal{A}}\mathcal{A} \cong$$



The Quiver

The Gabriel Quiver

The Gabriel quiver, $Q_{\mathcal{A}}$, of $\mathcal{A}$ is



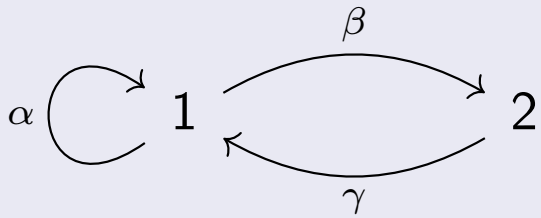
where the vertex i corresponds to the projective indecomposable, and therefore idempotent, P_i. Moreover, the admissible ideal I, such that $A \cong kQ/I$, is given by

$$I = \langle \alpha^2, \quad \gamma\beta, \quad \alpha\beta\gamma - \beta\gamma\alpha \rangle.$$

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Corollary (Erdmann)

The algebra $\mathcal{A}$ is of dihedral type and therefore symmetric, special biserial and of tame representation type.

String Algebras and Strings/Bands

Non-Projective Indecomposable Modules

Strings

$R_1^k : (\alpha\gamma^{-1}\beta^{-1})^k$ $R_1^{-k} : (\gamma^{-1}\beta^{-1}\alpha)^k$	$R_2^k : (\beta^{-1}\alpha\gamma^{-1})^k\beta^{-1}$ $R_2^{-k} : (\gamma^{-1}\beta^{-1}\alpha)^k\beta$	$R_3^k : \gamma(\alpha\gamma^{-1}\beta^{-1})^k$ $R_3^{-k} : (\gamma^{-1}\beta^{-1}\alpha)^k\gamma^{-1}$
	$B^k : (\gamma^{-1}\beta^{-1}\alpha)^k\gamma^{-1}\beta^{-1}$	
$C_1^k : (\beta^{-1}\alpha\gamma^{-1})^k$	$C_2^k : (\alpha\gamma^{-1}\beta^{-1})^k\alpha\beta$	$C_3^k : \gamma(\alpha\gamma^{-1}\beta^{-1})^k\alpha$
$D_1^k : (\beta^{-1}\alpha\gamma^{-1})^k\beta^{-1}\alpha\beta$	$D_2^k : (\alpha\gamma^{-1}\beta^{-1})^k\alpha$	$D_3^k : \gamma(\alpha\gamma^{-1}\beta^{-1})^k\alpha\gamma^{-1}$
$E_1^k : (\beta^{-1}\alpha\gamma^{-1})^k\beta^{-1}\alpha$	$E_2^k : (\alpha\gamma^{-1}\beta^{-1})^k\alpha\gamma^{-1}$	$E_3^k : \gamma(\alpha\gamma^{-1}\beta^{-1})^{k+1}\alpha\beta$

Table: Strings Modules, $k \geq 0$

Bands

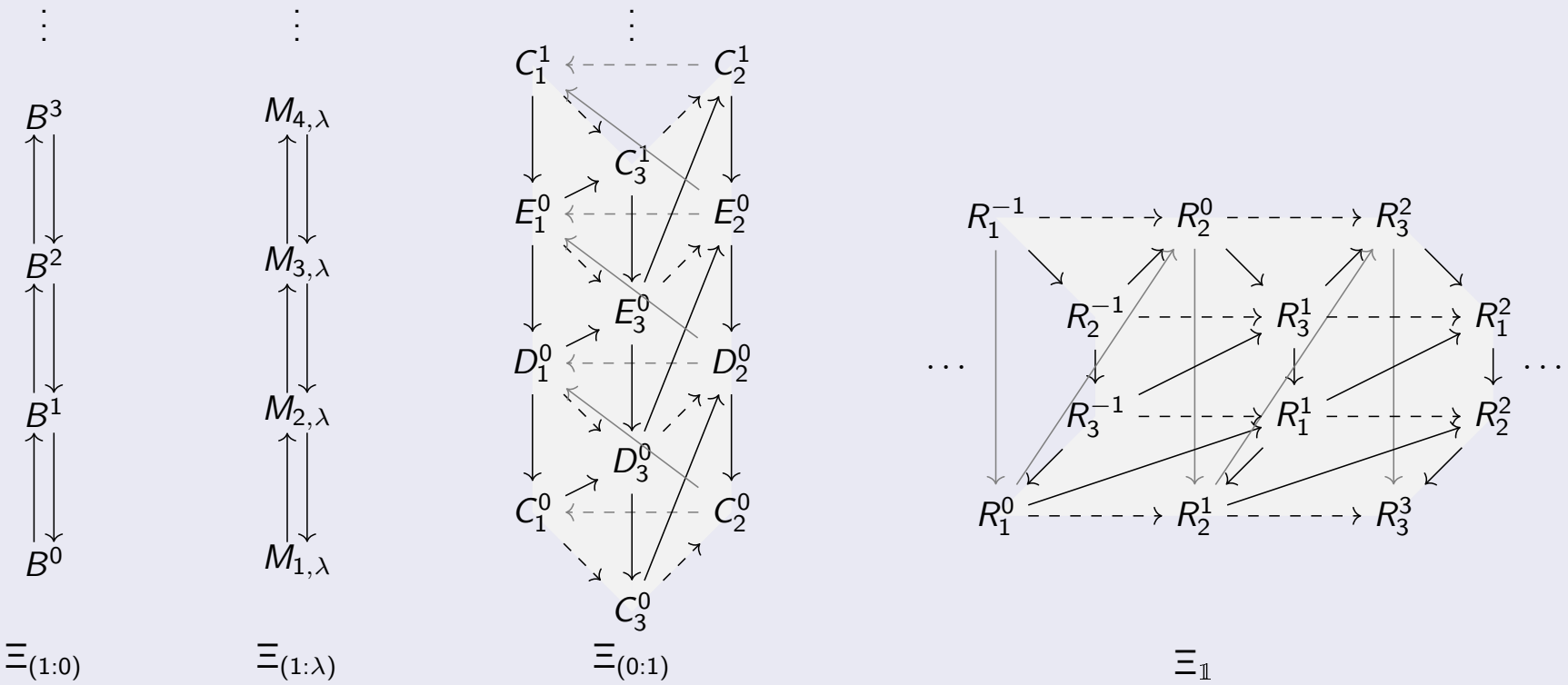
$M_{\lambda,n,w}$	$\lambda \in k^\times, \quad n \in \mathbb{N}, \quad w = \gamma^{-1}\beta^{-1}\alpha$

Table: Band Modules

AR Components of $\mathcal{A}$

Stable AR-quiver

The components are:



$\Xi_{(1:0)}$ $\Xi_{(1:\lambda)}$ $\Xi_{(0:1)}$ $\Xi_{\mathbb{1}}$

Figure: AR-components

Support for Finite Groups

Definition

Let G be a finite group and $H^*(G,k)$ it's cohomology ring. The **support variety** of a module M is

$$\mathcal{V}_G(M) := \{\mathfrak{p} \in \operatorname{Proj} H^*(G,k) \mid I_M \subseteq \mathfrak{p}\}$$

where $I_M := \operatorname{Ann}_{H^*(G,k)}(\operatorname{Ext}^*_{kG}(M,M)))$

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Example (Klein Four)

$kV_4 = k[s,t]/(s^2,t^2)$, $H^*(V_4,k) = k[u,v]$, $\mathfrak{p}_{(a:b)} = (au+bv)$

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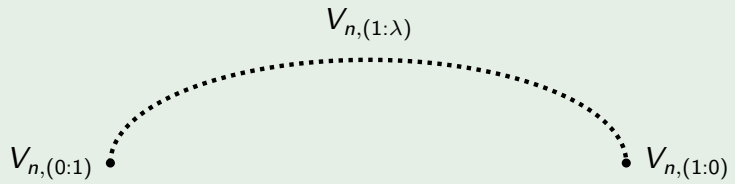
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$\cong \mathbb{P}^1_k$

Support for Finite Tensor Categories

Definition (Bergh-Plavnik-Witherspoon)

Let $\mathcal{C}$ be a finite tensor category and $H^*(\mathcal{C}) := \mathrm{Ext}^*_{\mathcal{C}}(\mathbb{1}, \mathbb{1})$ the cohomology algebra. The **support variety** of an object M is

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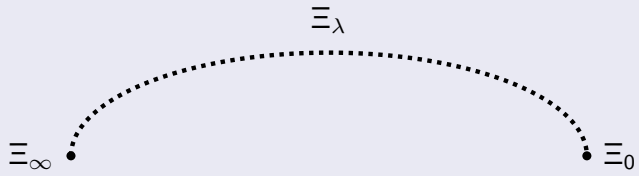
Properties

A good support theory has the following properties

1. $\mathcal{V}(M) = \emptyset$ if and only if M is projective
2. $\mathcal{V}(M \oplus N) = \mathcal{V}(M) \cup \mathcal{V}(M)$
3. $\mathcal{V}(M_i) \subseteq \mathcal{V}(M_j) \cup \mathcal{V}(M_k)$ for distinct $i, j, k \in \{1, 2, 3\}$ whenever $0 \to M_1 \to M_2 \to M_3 \to 0$ is a short exact sequence.
4. $\mathcal{V}(M \otimes N) = \mathcal{V}(M) \cap \mathcal{V}(N)$

Tensor Product Property

Support of Indecomposables

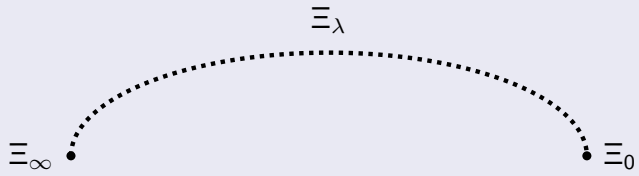


Theorem (Bergh-Plavnik-Witherspoon)

1. *For a periodic module* M, $\mathcal{V}(M) = \{\bullet\}$
2. *The following are equivalent*
   - $\mathcal{V}(M \otimes N) = \mathcal{V}(M) \cap \mathcal{V}(N)$ *for all objects* M, N *in* $\mathcal{C}$
   - $\mathcal{V}(M \otimes N) = \mathcal{V}(M) \cap \mathcal{V}(N)$ *for all* **indecomp periodic** M, N *in* $\mathrm{Ob}\,\mathcal{C}$.

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Theorem

The cohomological support $(\mathrm{Proj}\, H^*(Ver_{2^3}^+), \mathcal{V}(-))$ *satisfies the tensor product property.*

Balmer Spectrum

Definition

Let $\mathcal{C}$ be a symmetric tensor category.

1. $\operatorname{stab}\mathcal{C}$ is $\mathcal{C}$ "modulo" projective objects
2. A **TT-Category** $(\mathcal{T}, \otimes, \oplus)$ is a symmetric monoidal triangulated category
3. A **prime tensor ideal** $\mathcal{P} \subseteq \mathcal{T}$ is a thick tensor ideal satisfying $a \otimes b \in \mathcal{P} \implies a \in \mathcal{P}$ or $b \in \mathcal{P}$.
4. The **Balmer spectrum** $\operatorname{Spc}(\mathcal{T})$ is the set of prime tensor ideals of $\mathcal{T}$

Spectra

Spec of a ring R		Spc of $\operatorname{stab}\mathcal{C}$
R a commutative ring	$\rightsquigarrow$	$\operatorname{stab}\mathcal{C}$ a tensor triangulated category
$\operatorname{Proj} R$ the prime ideals of R	$\rightsquigarrow$	$\operatorname{Spc}(\operatorname{stab}\mathcal{C})$ the prime tensor ideals of $\operatorname{stab}\mathcal{C}$

Spectrum of $\mathrm{Ver}_{2^3}^+$

Theorem

The comparison map

$$\mathrm{Spc}(\mathrm{stab}\, Ver_{2^3}^+) \xrightarrow{\rho} \mathrm{Proj}\, H^*(Ver_{2^3}^+)$$

is a homeomorphism

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Prime Tensor Ideals

For each $\mathfrak{p} \in \mathrm{Proj}\, H^*(\mathrm{Ver}_{2^3}^+)$,

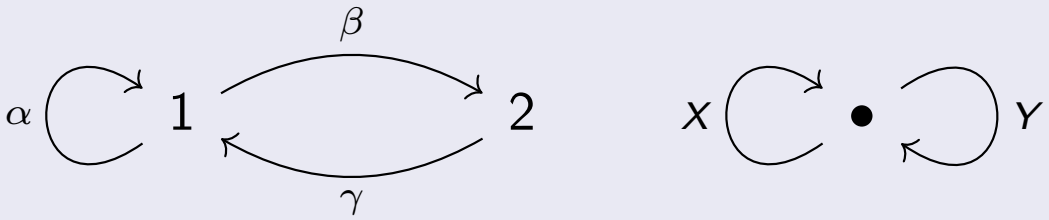
$$\mathcal{P}_{\mathfrak{p}} = \mathrm{thick}\left(\bigcup_{\mathfrak{q}\neq\mathfrak{p}} \Xi_{\mathfrak{q}}\right) \qquad \mathcal{P}_{(0)} = \mathrm{thick}\left(\bigcup_{\mathfrak{p}} \Xi_{\mathfrak{p}}\right)$$

are the only prime tensor ideals.

Klein Four as Corner Sublgebra

Restriction/(Co)Induction

There is a natural inclusion of $\mathcal{B} := k[s,t]/(s^2,t^2) \cong e_1\mathcal{A}e_1 \subseteq \mathcal{A}$ to our algebra $\mathcal{A}$



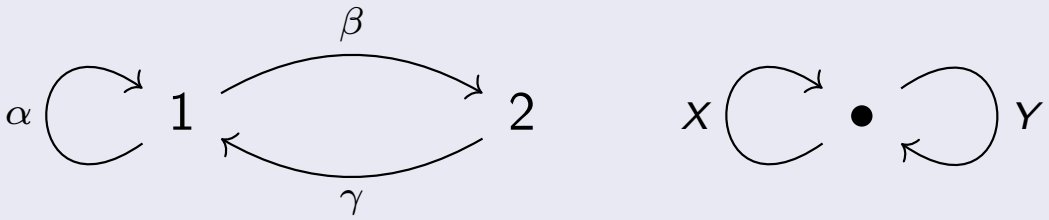
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$$\mathcal{A}-\text{mod} \underset{T_e, L_e}{\overset{\text{res}_e}{\rightleftarrows}} \mathcal{B}-\text{mod}$$

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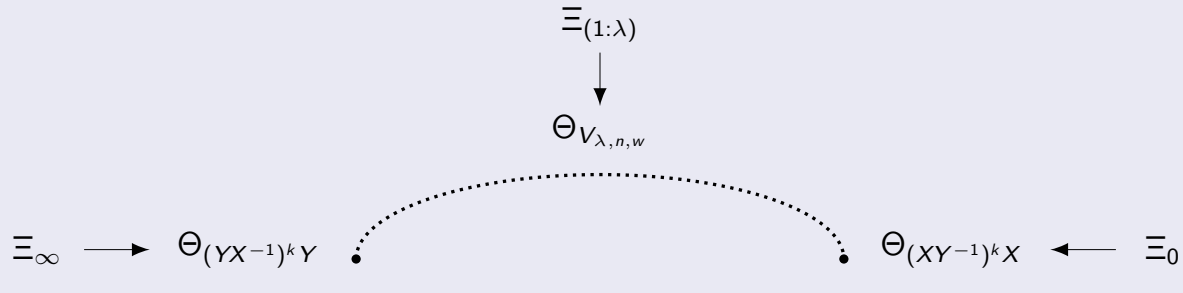
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Strings, Bands and Components

1. Band modules $V_{\lambda,n,w}$ where $w = XY^{-1}$, $n \in \mathbb{N}$ and $\lambda \in k^\times$
2. Strings $(XY^{-1})^k X$, $(YX^{-1})^k Y$, $(XY^{-1})^k$, $(Y^{-1}X)^k$, $k \geq 0$

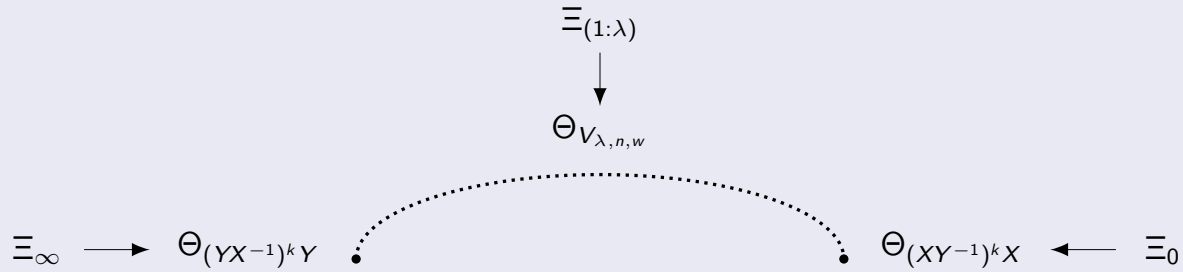
Klein Four as Corner Subalgebra

Restriction and Components



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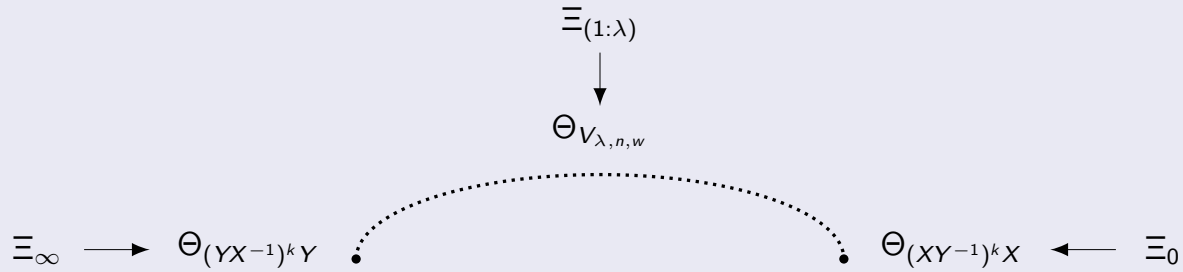
Proposition

There is a lattice isomorphism

$$\mathrm{Thick}(\mathrm{Stab}\,\mathcal{A} - \mathit{mod}) \overset{\simeq}{\rightleftarrows} \mathrm{Thick}(\mathrm{Stab}\,\mathcal{B} - \mathit{mod})$$

Klein Four as Corner Subalgebra

Restriction and Components



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Proposition

The restriction functor induces a map $\varphi : H^*(Ver_{2^3}^+) \to H^*(\mathcal{B})$ *which induces a homeomorphism*

$$\overline{\varphi^*} : \mathrm{Proj}(H^*(\mathcal{B})) \xrightarrow{\simeq} \mathrm{Proj}(H^*(Ver_{2^3}^+)).$$

Conclusion

Thank You!